

Retrospective US inflation adjustment: compact forecasting evidence and limits

A two-term partial-adjustment equation improves on persistence in development and confirmation, but does not beat the training-only constant in the reserved recent period. The portfolio supports an interpretable scoped forecasting relationship, not a new structural inflation law.

Data and evaluation

Six US seasonally adjusted CPI/CES series from the preserved BLS bulk snapshot. Development uses 1990-2022, with forward five-year validation blocks from 2000; confirmation uses 2023-2025 September, 33 months in 11 quarters. Complete-calendar eligibility excludes later months after a missing 2025 October CPI observation. The source snapshot extends into 2026 but those later dates are not silently imputed.

Target: **Next-month seasonally adjusted all-items CPI log change**, measured as percentage points per month (100 natural log index ratio). Primary error: percentage points. Only calendar months through t-1 enter predictors for month t. This is a retrospectively revised and seasonally adjusted bulk snapshot, not an as-issued forecast archive. Calendar lags do not establish historical release availability; the nominal origin is after prior-month releases. No month-t prices/employment or CPI accounting decomposition enters inputs.

Whole calendar quarter; forward chronological development folds and 2023-onward confirmation. One national economic system, overlapping lag windows and persistent macro shocks; quarters are not independent economies. All regression coefficients, centering/scales and flexible centers fit only preceding development blocks. No held-quarter target fitting.

Findings and equations

$$\hat{p} = p_{12} + a(p_1 - p_{12}) + b[(h + m)/2 - p_1]$$

The notation and all fitted coefficients are defined below.

P100-016-F01 - reproduction (partially_supported). Lagged partial adjustment provides an interpretable predictor better than persistence, with limited cross-period superiority.

$$p_{\text{hat}} = p_{12} + a(p_1 - p_{12}) + b[(h + m)/2 - p_1]$$

Partial persistence and sector-relative price adjustment are established forecasting motifs. The coefficient is phenomenological, not an identified causal relaxation constant.

Falsifying evidence and limits: Confirmation constant MAE 0.105693 beats reference 0.108585; uncertainty is group descriptive, no independent-quarter confidence.

P100-016-F02 - informative_falsification (supported). Adding employment pressure, shock asymmetry, robust fitting or acceleration does not establish a new stable inflation mechanism.

See frozen attempts 003-006 and nested controls.

Employment improved a worst-quarter diagnostic while damaging mean transfer; asymmetry traded mean and worst-group performance.

Falsifying evidence and limits: No causal intervention, revision-free vintages, or independent economic systems; no structural law or demonstrated financial value.

Selected executable model: attempt_002. Let p_1 , p_{12} denote previous-month inflation and the trailing 12-month mean, h , m previous housing and medical inflation. The selected equation is $p_{\text{hat}} = p_{12} + a(p_1 - p_{12}) + b[(h + m)/2 - p_1]$, where $a = 0.6788581552753963$, $b = 0.25333515480765123$. Every term

is in monthly log-change percentage points. This is a lagged adjustment representation; housing/medical averages are not an exact CPI decomposition.

All coefficients, input definitions, training groups and transformations are in `rules.json`; the runnable reference performs no fitting.

Compact numerical evidence

Errors are computed within declared complete groups (with final survey weights when supplied), then averaged equally across groups.

Frozen model	Development error	Confirmation error	Worst confirmation group
reference	0.2006199	0.1085852	0.2049382
annual_mean	0.224154	0.1167382	0.2012013
constant	0.2156496	0.1056932	0.177262
flexible	0.2260805	0.1132284	0.2328886
persistence	0.2295867	0.1289942	0.3072

Confirmation contains 33 scored observations in 11 groups. Individual-group scores and every attempted model remain available. Reference confirmation MAE 0.108585 percentage points versus persistence 0.128994, annual mean 0.116738, flexible 0.113228 and constant 0.105693. The constant is better, so universal predictive superiority is falsified. No post-confirmation reselection: attempt 005/006 remain diagnostics despite some lower recent errors.

Interpretation, limitations and use

1990-2026 US selected national seasonally adjusted series. Revised historical values and seasonal factors can incorporate later information. No causal Phillips-curve, policy intervention, operational financial-return or structural-inflation law claim. Quarter-group errors/ranges; no row bootstrap or independent-quarter population confidence. Confirmation will become exposed.

Useful as a compact source-aware retrospective forecasting reference and a test of avoiding policy/causal overclaim. Revised seasonal factors and benchmarked employment make this unsuitable as a proven real-time trading or monetary-policy model.

Reproducibility and prior art

Run `python run.py` to verify hashes and replay every frozen equation. Predictions use declared input columns only. The shared benchmark evaluator can score alternative equations under the same task; numerical agreement with these coefficients is not required. Public source-aware confirmation is now exposed. No independent experimental replication or certified novelty is claimed.

- [Annual CPI seasonal re-estimation revises prior five years; 2021-2025 recalculated in February 2026.](#)
(primary source checked).

- [CES revisions reflect later responses and annual benchmarks; seasonal histories can change five years.](#)
(primary source checked).